

$$1. \lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n}$$

$$2. f(x) = \frac{1 - e^{\frac{1}{x}}}{1 + e^{\frac{1}{x}}} \arctan \frac{1}{x}$$

$$3. f(x) = \arctan x - x$$

已知 $a_{n+1} = \sqrt{2 + a_n}$ 注意括号 $a_1 = \sqrt{2}$, 求证: a_n 收敛

4.

$$\text{五. 计算 } \lim_{x \rightarrow \infty} \left(\frac{2x+1}{2x+3} \right)^x$$

$$\text{六: } a, b \text{ 为何值时 } f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+1} + ax^2 + bx}{x^{2n} + 1} \quad (-\infty < x < +\infty) \text{ 连续}$$

$$7. \text{ 讨论 } f(x) = \begin{cases} x^2 \arctan \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases} \text{ 在 } x=0 \text{ 处是否可导}$$

$$8. \begin{cases} x = f'(t) \\ y = t f'(t) - f(t) \end{cases} \quad f''(t) \text{ 存在且不为 } 0. \text{ 求 } \frac{dy}{dx}$$

$$9. \text{ 设 } y = y(x) \text{ 由 } e^y + xy = e \text{ 确定.}$$

十. 利用单调性证明: 当 $x \in [\frac{1}{2}, 1]$ 时, $\arctan x - \ln(1+x^2) \geq \frac{\pi}{4} - \ln 2$

证明: 令 $f(x) = \arctan x - \ln(1+x^2) - \frac{\pi}{4} + \ln 2$

十一. 设 $f(x) = \begin{cases} x^{2x} & x > 0 \\ x+1 & x \leq 0 \end{cases}$ 求 $f(x)$ 的极值

十二. 设 $f(x)$ 在 $[a, b]$ 连续 在 (a, b) 可导 且 $f(a) = f(b) = 0$. 证明:

$\exists \xi \in (a, b)$, 使 $f'(\xi) = -\lambda f(\xi)$ 这里 λ 为 $\forall$ 实数.

1. $\lim_{n \rightarrow \infty} \frac{2^{n+1} + 3^{n+1}}{2^n + 3^n} = \lim_{n \rightarrow \infty} \frac{\left(\frac{2}{3}\right)^{n+1} + 1}{\frac{1}{3}\left(\frac{2}{3}\right)^n + \frac{1}{3}} = 3$

2. $f(x) = \frac{1 - e^{\frac{1}{x}}}{1 + e^{\frac{1}{x}}} \arctan\left(\frac{1}{x}\right) \quad x=0$

$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{e^{-\frac{1}{x}} - 1}{e^{-\frac{1}{x}} + 1} \arctan\left(\frac{1}{x}\right) = -\frac{\pi}{2}$ 可去间断点

$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1 - e^{\frac{1}{x}}}{1 + e^{\frac{1}{x}}} \lim_{x \rightarrow 0^-} \arctan \frac{1}{x} = 1 \cdot \left(-\frac{\pi}{2}\right) = -\frac{\pi}{2}$

3. $f(x) = \frac{\arctan x - x}{1+x^2}$ 单调区间为 $(-\infty, +\infty)$

$f'(x) = \frac{1}{1+x^2} - 1 = \frac{-x^2}{1+x^2} \leq 0$

4. 显然有 $A_n > 0$

A_n

$A_{n+1} = \sqrt{2 + a_n}$

数列有上界

$\Rightarrow$

下面用[数学归纳法](#)证明[单调性](#)

$[A_{n+1}]^2 = 2 + A_n$

1) 显然: $A_2 > A_1$

$\Rightarrow$

2) 假设 $A_n > A_{n-1}$

$[A_{n+1} - 2][A_{n+1} + 2] = A_n - 2$

$\Rightarrow$

其中:

$A_{n+1} = \sqrt{2 + A_n} > \sqrt{2 + A_{n-1}} = A_n$

$A_n > 0$

由 1)、2) 可得 $A_{n+1} > A_n$

$\Rightarrow$

$\Rightarrow$

$A_{n+1} + 2 > 0$

A_n 为递增数列

$\Rightarrow$

综合有上界和递增性, 根据单调有界准则可知 A_n 收敛

$A_{n+1} - 2$ 与 $A_n - 2$ 同号

再求极限得 $\lim_{n \rightarrow \infty} a_n = 2$

又 $A_1 = \sqrt{2}$

例2. 2. 对 $\forall n \geq 1$ $X_n < 2$ (1) $n=1$ $X_1 = \sqrt{2} < 2$ (2) 若 $n=k$ 时结论成立 即 $X_k < 2$
 2.1 当 $n=k+1$ 时 $X_{k+1} = \sqrt{2+X_k} < 2$ $\therefore \lim_{n \rightarrow \infty} (X_n) = A = 2$
 $\therefore X_{n+1} = \sqrt{2+X_n}$ 令 $y_n = X_{n+1}$ $z_n = \sqrt{2+X_n} \rightarrow \sqrt{2+A}$ $y_n = z_n$
 $\therefore \lim_{n \rightarrow \infty} y_n = A = \lim_{n \rightarrow \infty} z_n = \sqrt{2+A}$
 $\Rightarrow A = \sqrt{2+A} \Rightarrow A^2 = 2+A \Rightarrow A^2 - A - 2 = 0 \Rightarrow A = 2$ 或 $A = -1$

五. 计算 $\lim_{x \rightarrow \infty} \left(\frac{2x+1}{2x+3} \right)^x$ 1^∞
 $= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{-2}{2x+3} \right)^{\frac{2x+3}{-2}} \right]^{\frac{-2x}{2x+3}} \rightarrow -1$
 $= e^{-1} = \frac{1}{e}$
 $\lim_{n \rightarrow \infty} \left(\frac{n + \ln n}{n - \ln n} \right)^{\frac{n}{\ln n}}$ 1^∞
 $\frac{1 + \frac{\ln n}{n}}{1 - \frac{\ln n}{n}}$

六. a, b 为何值时 $f(x) = \lim_{n \rightarrow \infty} \frac{x^{2n+1} + ax^2 + bx}{x^{2n} + 1}$ $(-\infty < x < +\infty)$ 连续
 解: $f(x) = \begin{cases} \frac{1}{x} & |x| > 1 \\ \frac{1+a+b}{2} & |x| < 1 \\ \frac{a-b-1}{2} & x = 1 \\ \frac{a-b-1}{2} & x = -1 \end{cases}$
 $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = \frac{1+a+b}{2} \Leftrightarrow 1 = \frac{a+b}{2} = \frac{1+a+b}{2} \Rightarrow \begin{cases} a=0 \\ b=1 \end{cases}$
 $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = \frac{a-b-1}{2} \Leftrightarrow -1 = \frac{a-b-1}{2}$

7. 讨论 $f(x) = \begin{cases} x^2 \arctan \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$ 在 $x=0$ 处是否可导
 解: $f'(0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x) - f(0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^2 \arctan \frac{1}{\Delta x}}{\Delta x} = 0$
 $\therefore x=0$ 处可导

8. $\begin{cases} x = f'(t) \\ y = tf'(t) - f(t) \end{cases}$ $f''(t)$ 存在且不为 0. 求 $\frac{dy}{dx}$

解: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{f'(t) + tf''(t) - f'(t)}{f''(t)} = t$
 $\frac{dy}{dx} = \frac{d}{dx}(\frac{dy}{dt}) = \frac{d}{dx}(t) = \frac{d}{dt}(t) / \frac{dx}{dt} = \frac{1}{f''(t)}$

9. 设 $y = y(x)$ 由 $e^y + xy = e$ 确定. 求 $\frac{d^2y}{dx^2}$

解: 方程两边关于 x 求导得: $e^y \cdot y'(x) + y + x \cdot y' = 0 \Rightarrow y' = \frac{-y}{e^y + x} = \frac{dy}{dx}$

$\frac{d^2y}{dx^2} = \frac{d}{dx}(\frac{dy}{dx}) = \frac{d}{dx}\left(\frac{-y}{e^y + x}\right) = \frac{-y'(x)(e^y + x) + y(e^y y'(x) + 1)}{(e^y + x)^2}$

$= \frac{\frac{y}{e^y + x}(e^y + x) + y(e^y \frac{-y}{e^y + x} + 1)}{(e^y + x)^2} = \frac{2y - \frac{e^y y^2}{e^y + x}}{(e^y + x)^2} = \frac{2y(e^y + x) - e^y y^2}{(e^y + x)^3}$

十. 利用单调性证明: 当 $x \in [\frac{1}{2}, 1]$ 时 $\arctan x - \ln(1+x^2) \geq \frac{\pi}{4} - \ln 2$

证明: 令 $f(x) = \arctan x - \ln(1+x^2) - \frac{\pi}{4} + \ln 2$. $\therefore f(\frac{1}{2}) = \arctan \frac{1}{2} - \ln(1+\frac{1}{4}) - \frac{\pi}{4} + \ln 2$

$f(1) = 0$. $f'(x) = \frac{1}{1+x^2} - \frac{2x}{1+x^2} = \frac{1-2x}{1+x^2}$ 当 $x \in (\frac{1}{2}, 1)$ 时, $f'(x) < 0$

$\therefore f(x)$ 在 $[\frac{1}{2}, 1]$ 上单调减. $\therefore$ 对 $\forall x \in [\frac{1}{2}, 1]$ $f(x) \geq f(1) = 0$ 即 $\arctan x - \ln(1+x^2) \geq \frac{\pi}{4} - \ln 2$ 得证

十一. 设 $f(x) = \begin{cases} x^{2x} & x > 0 \\ x+1 & x \leq 0 \end{cases}$ 求 $f(x)$ 的极值

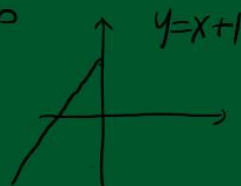
证明: $\lim_{x \rightarrow 0^+} x^{2x} = \lim_{x \rightarrow 0^+} e^{2x \ln x} = e^{\lim_{x \rightarrow 0^+} 2x \ln x} = e^0 = 1$

$\therefore f(x)$ 在 0 点连续. 当 $x > 0$ 时 $f'(x) = (e^{2x \ln x})' = x^{2x}(2 \ln x + 2x \cdot \frac{1}{x}) = 2x^{2x}(\ln x + 1)$

当 $x = \frac{1}{e}$ 时 $f'(x) = 0$ 当 $x \in (0, \frac{1}{e})$ 时 $f'(x) < 0$ 而当 $x > \frac{1}{e}$ 时 $f'(x) > 0$

$\therefore x = \frac{1}{e}$ 为极大值点 极大值 $f(\frac{1}{e}) = 1$

当 $x > \frac{1}{e}$ 时 $f'(x) > 0$ $\therefore x = \frac{1}{e}$ 为极小值点. 极小值为: $(\frac{1}{e})^{\frac{2}{e}}$



十二. 设 $f(x)$ 在 $[a, b]$ 连续 在 (a, b) 可导 且 $f(a) = f(b) = 0$.

$\exists \xi \in (a, b)$ 使 $f'(\xi) = -\lambda f(\xi)$ 这里 λ 为 $\forall$ 实数.

证明: 令 $\varphi(x) = e^{\lambda x} \cdot f(x)$ $\therefore \varphi(x)$ 在 $[a, b]$ 连续 在 (a, b) 可导

$\Rightarrow \varphi(a) = \varphi(b) = 0$ $\therefore$ 由罗尔定理 $\exists \xi \in (a, b)$ 使 $\varphi'(\xi) = 0$

$\varphi'(\xi) = \lambda e^{\lambda \xi} f(\xi) + e^{\lambda \xi} f'(\xi) = e^{\lambda \xi} (\lambda f(\xi) + f'(\xi)) = 0$

$\Rightarrow \lambda f(\xi) + f'(\xi) = 0$

即 $f'(\xi) = -\lambda f(\xi)$

$\therefore$ 得证

$\varphi'(x) = \lambda e^{\lambda x} f(x) + e^{\lambda x} f'(x)$